

Civics Group	Index Number	Name (use BLOCK LETTERS)
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**ST ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIMS**

H1 BIOLOGY

8876/2

Paper 2: Structured & Essay Questions

[MARK SCHEME]

Monday

10 Sept 2018

2 hours

Additional Materials: Answer Paper
Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/13
2	/12
3	/9
4	/11
Sub-total	/45
5/6	/15
Total	/60

Conceptual error (C)	Expression problem (E)	Misreading the question (Q)	Data quoting problem (D)

This document consists of **XX** printed pages.

[Turn over]

Section A

Answer **all** questions.

QUESTION 1

Fig. 1.1 shows Process X in an eukaryotic cell which produces ribosomal RNA (rRNA).

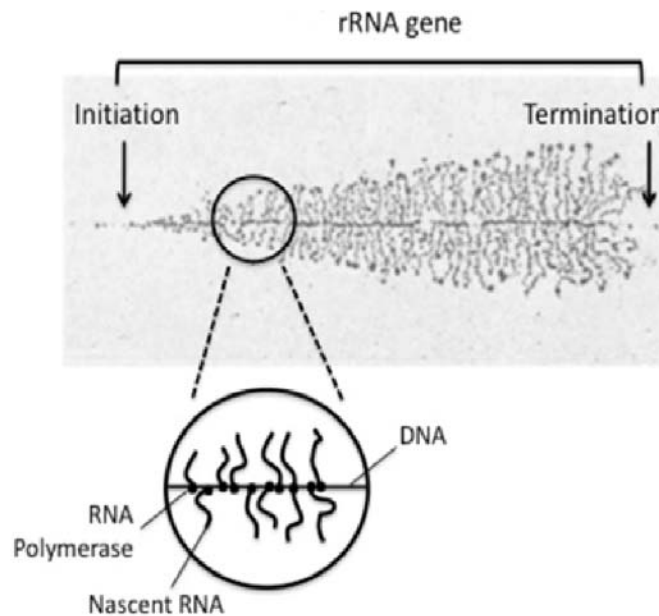


Fig. 1.1

(a)(i) Name the Process X occurring in **Fig. 1.1**.

.....[1]
1. Transcription

(ii) List one molecule **not mentioned in Fig. 1.1** that is required for Process X.

.....[1]
1. General transcription factor (**Reject:** Specific transcription factor due to it not being a real/essential requirement for transcription)
/ ribonucleotides
/ transcription initiation factors;

(iii) Suggest how RNA polymerase is able to recognise and bind to the promoter on DNA and not to other DNA regions.

-[2]
1. RNA polymerase contains a **DNA-binding site/domain** [Reject: active site] which recognize and bind to specific DNA sequence in the **promoter**;
 2. ref. **Nucleotide sequence** / length / major and minor grooves of promoter offers a **complementary shape** to DNA-binding site/domain of RNA polymerase; [Reject: complementary base pairing]

(iv) Explain for the observed pattern of Process X in **Fig. 1.1**.

-[2]
1. [Describe] **Shorter** RNA transcripts seen at the **beginning** of the DNA template strand, which get **longer** till the **end** of the transcription unit, (where the transcripts detach from the DNA template after transcription termination)
 2. [Explain] Due to simultaneous transcription of rRNA gene **by multiple RNA polymerases**, causing RNA transcripts to extend perpendicularly from DNA template strand ;

(v) State the roles of rRNA in protein synthesis.

-[2]
1. The rRNA in ribosomes holds the tRNA and mRNA together in **close proximity**, (via complementary base pairing / hydrogen bonds)
 2. positions the new amino acid for addition to the carboxyl end of the growing polypeptide
 3. rRNA peptidyl transferase activity catalyzes formation of a peptide bond between the new amino acid and the polypeptide chain
 4. Ref. rRNA associate with proteins to form ribosomal subunits / ribosomes (which synthesizes proteins)

(b) During protein synthesis in cells of an embryo, all tRNA molecules with UAC anticodon sequence, are observed to be bound to the arginine amino acid instead of methionine.

(i) Suggest how these tRNA molecules attached with the wrong amino acid might arise.

.....[2]

1. Ref. possible mutation in the **gene** sequence for the aminoacyl tRNA synthetases,
2. resulting in **altered 3D conformation** of active site which is **complementary** (in shape) to the amino acid arginine and the corresponding tRNA with anticodon UAC

(ii) Suggest and explain the effect of this wrong pairing of amino acid to tRNA on the embryo.

.....[3]

1. ref. altered **primary sequence** of polypeptides (all methionine replaced by arginine) and folding of polypeptides to **tertiary structure** / 3D conformation is affected;
2. ref. **non-functional proteins** made in cells
3. ref. possible disruption of metabolic processes in the **cell** / cells might die easily, **embryo cannot further develop** into a fetus

[Total: 13]

QUESTION 2

(a) About one third of the injuries to racehorses involve tendon damage.

In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells (usually around 10 million cells for one tendon) were then injected into its damaged tendons.

80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

(i) Explain how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing the tendon injuries.

.....[2]

- 1 ref. bone marrow stem cells as **multipotent**;
/ able to differentiate into cells of **same/specific lineage**;
- 2 ref. bone marrow stem cells able to **respond to specific molecular signals** to differentiate;
- 3 AVP: stem cells containing a complete set of genome;

(ii) Suggest an advantage of the above stem cell therapy.

.....[1]

- 1 Eliminates the risk of tissue/donor rejection if the cells were taken from another donor.
- 2 ref. stem cells **capable of continual self-renewal** and can be used to **expand cell numbers** in vitro / in the lab;

(iii) Similar to stem cells, cancer cells are also capable to dividing many times via mitosis. List 2 differences between stem cells and cancer cells.

.....[2]

	Stem cells	Cancer cells
1	Differentiate into specialised cell types;	Do not differentiate into specialised cell types / remain undifferentiated and unspecialised;
2	Experience contact inhibition;	Does not experience contact inhibition;
3	Experience anchorage dependence / does not metastasize;	Does not experience anchorage dependence / metastasize to set up secondary tumors;
4	Divides or stops dividing upon response to molecular signals	Divides indefinitely / does not respond to molecular signals;
5	AVP: i.e. normal proto-oncogene / tumor suppressor gene	AVP: i.e. mutated proto-oncogene / tumor suppressor gene

(b) In a dihybrid inheritance, gene B/b codes for flower colour while gene H/h codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then undergo a test cross.

(i) Predict the expected phenotypic ratio in the F2 progeny.

-[1]
- 1 1 Red flower, fan-shaped leaf : 1 Red flower, oval leaf : 1 Yellow flower, fan-shaped leaf : 1 Yellow flower, oval leaf

[Reject: 1:1:1:1 with no phenotypes]

(ii) Identify and explain the dominant traits in this dihybrid cross.

-[2]
- 1 red flowers and fan-shaped leaves;
- 2 ref. **dominant traits/alleles** being expressed in the **phenotype** in the **heterozygous** condition (i.e. F1 progeny)
- / **recessive alleles** (coding for yellow flowers and fan-shaped leaves), **being masked by dominant alleles** in the **heterozygous** condition;

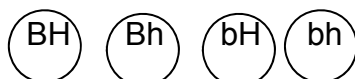
(iii) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross if inheritance is Mendelian.

.....[3]

F₁ phenotypes: Red flower, x Yellow flower,
 Fan-shaped leaf Oval leaf

F₁ genotype: BbHh bbhh

F₁ gametes



F₂ genotypes:

Punnett square:

	BH	Bh	bH	bh
bh	BbHh (Red flower, Fan-shaped leaf)	Bbhh (Red flower, Oval leaf)	bbHh (Yellow flower, Fan-shaped leaf)	bbhh (Yellow flower, oval leaf)

F₂ / Progeny
phenotypes:

Red flower, Fan-
shaped leaf

Red flower,
Oval leaf

Yellow flower,
Fan-shaped leaf

Yellow flower,
oval leaf

F₂ / Progeny
phenotypic ratio:

1

1

1

1

Mark scheme:

1. F₁ phenotype and genotypes
2. Parental gametes – (Gametes **must** be circled)
3. F₂ genotypes correspond to phenotypes

(iv) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

.....[1]

1. Humans produce **limited offspring**. (This makes statistical tests difficult)
2. Humans have a long life span and **some traits only appear at a later stage in life**

[**Reject:** some human traits are under continuous variation. Plants also have continuous variation]

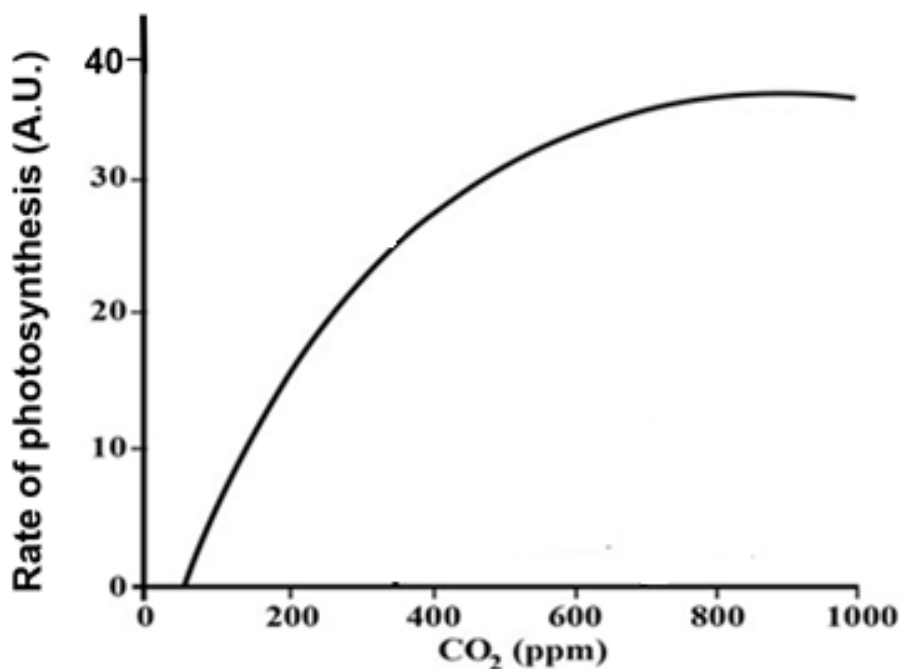
[Total: 12]

QUESTION 3

(a) In response to climate change, the rising carbon dioxide level has affected almost all crucial biological processes, including photosynthesis, respiration, and antioxidant systems, as well as other key secondary metabolisms in plants.

An experiment was designed to measure the rate of photosynthesis at a range of carbon dioxide concentrations provided to the plant.

The following results were obtained:



(i) Explain the experimental results obtained in relation to CO₂ being a limiting factor of photosynthesis.

- [2]
- 1 ref. carbon dioxide being a limiting factor of photosynthesis at values **lower than 700ppm** (Accept range: 700-800ppm)
/At higher concentrations of carbon dioxide tested (i.e. 700-800ppm and above), the rate of photosynthesis is affected by other limiting factors / carbon dioxide is no longer a limiting factor;
 - 2 [Evidence] rate of photosynthesis reaches a **plateau /remains constant around 35-37AU**

(ii) Suggest why the rate of photosynthesis reaches a plateau after a while, given that sufficient light was provided.

- [1]
- 1 Rate of PS limited by **amount of Rubisco** in the plant;

(iii) NADP^+ is an important electron carrier found in plants, but its level decreases sharply in the day. Suggest the significance of the decrease of NADP^+ during the day.

..... [2]

(ref. light-dependent reactions during day

/ photoexcited electrons (from PS I and II)being passed down ETC)

- 2 NADP^+ acts as the **final electron acceptor**, forming NADPH ;
/ NADP^+ **accepts electrons and H^+ ions** to form NADPH ;
- 3 NADPH (and ATP) used in **Calvin cycle** / light independent reactions to reduce PGA to PGAL **to synthesis of glucose**;

(b) Central to the energy metabolism of aerobically respiring cells is the enzyme ATP synthase. It consists of multiple subunits, coded for by several genes.

The enzyme is located on the inner mitochondrial membrane and catalyses the reversible reaction of ATP synthesis from ADP in intact mitochondria. Here, the **proton motive force** is required to drive the ATP synthesis.

Explain how the structure, including protein components, of the inner mitochondrial membrane is significant in driving the reaction of ATP synthase towards ATP synthesis during aerobic respiration.

.....[4]

[Important structure – hydrophobic fatty acid tails]

- 1 **Hydrophobic** fatty acid tails / hydrocarbon chains / hydrophobic core of inner mitochondrial membrane **repels / does not allow hydrophilic / charged H^+ ions** to pass through membrane;
- 2 Allows **proton gradient / proton motive force** to be established;

[Important composition – series of electron carriers]

- 3 Electrons passed down a series of **electron carriers** / **ETC** present on membrane with increasing electronegativity and in order of **decreasing energy levels**; (until they reach final electron acceptor – oxygen)
- 4 Energy released during transfer of electrons along series of electron carriers used to **pump H^+ ions from mitochondrial matrix into intermembrane space**;

[Important composition – ATP synthase]

- 5 **ATP synthase** embedded in an orientation that allows facilitated **diffusion of H^+ ions from intermembrane space to mitochondrial matrix** to be coupled with ATP synthesis;

[Total: 9]

QUESTION 4

(a) Climate change, in the form of global warming, is expected to have an impact on various organisms. This impact however, varies geographically, particularly for insects.

Fig. 4.1 and **Fig. 4.2** below show the fitness curves of representative insects from temperate and tropical locations respectively.

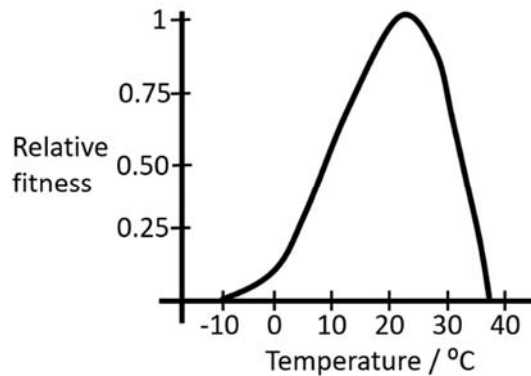


Fig. 4.1: Fitness curve of representative insect from temperate location

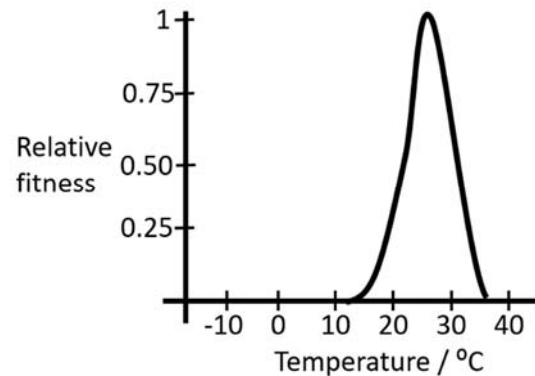


Fig. 4.2: Fitness curve of representative insect from tropical location

The mean annual temperature of the temperate location is 11°C, with a typical temperature range from 1°C to 20°C.

On the other hand, the mean annual temperature of the tropical location is 27°C, with a typical temperature range from 21°C to 31°C.

(i) Suggest what is meant by the term relative fitness in **Fig. 4.1** and **Fig. 4.2**.

-[2]
- 1 **[Fitness]** a measure of evolutionary success as indicated by the number of surviving offspring left to produce the next generation;
 - 2 **[Relative]** with reference to / as compared to **fitness at optimum temperature** (relative fitness of 1);

- (ii) With reference to **Fig. 4.1** and **Fig. 4.2** and the information given, predict and explain which insects, from the temperate or tropical location, would face a greater extinction risk as a result of global warming, assuming the warming elevates temperatures equally at both locations.

[4]

.....[4]
[Predict]

- 1 Insects from the **tropical location**;

[Explain]

- 2 As these insects are already living **close to / just under / around / (sometimes) beyond** their **optimum temperature** for maximum relative fitness of 1 / these insects have a **narrower temperature tolerance**;
- 3 [Quote] Optimum temperature = 26 - 28°C (Accept: any figure within range) + mean annual temperature of the tropical location is 27°C OR typical temperature range from 21 to 31°C;
- 4 Ref. temperature predicted to increase by 2 – 3 °C as a result of global warming;
- 5 Ref. lower relative fitness beyond optimum temperature / Ref. temperature increase likely to lower relative fitness, thus, increasing extinction risk;
- 6 Ref. temperature increase likely to increase relative fitness of insects from temperate location instead, reducing extinction risk;

- (iii) Suggest one strategy that the insects in (ii) can employ to reduce the impact of global warming on themselves.

[1]

- 1 Migration to cooler climate / regions / habitats / higher altitude / latitude;

(b) Carbon dioxide is one of the greenhouse gases that contribute to global warming and subsequently climate change.

A group of high school students decided to test whether varying temperatures would correspondingly affect the mean carbon dioxide gas emission.

Table 4.1 shows the mean carbon dioxide gas emission after exposing a fixed number of mealworms to different temperatures. Carbon dioxide gas emission was measured before and after exposure to experimental temperature.

Table 4.1 showing effects of varying temperatures on mean carbon dioxide gas emission, measured in parts per million (ppm).

Temperature / °C	Mean Carbon dioxide gas emission / ppm	
	Before exposure to experimental temperature	After exposure to experimental temperature
30.0	445 ± 25	450 ± 17
40.0	450 ± 20	500 ± 30
50.0	460 ± 17	540 ± 18

Values represent mean ± standard deviation.

(i) Describe the patterns shown by the data in **Table 4.1**.

.....[2]

- [Trend] The mean carbon dioxide gas emission increases as temperature increases;
- [Quote data] As temperature increases from 30.0°C to 50.0°C, emission increases from 450ppm to 540ppm.
OR
- [Trend] **Increase** in carbon dioxide gas emission **after exposure** to experimental temperatures **compared to before** exposure, gets bigger as temperature increases
- [Quote data] As temperature increases from 30.0°C to 50.0°C, increase in emission increases from 5ppm to 80ppm

(c) Corals are affected by rising temperatures in ocean waters. Explain how.

.....[2]

- Ref. Absorption of **more carbon dioxide which dissolves** when ocean waters get warmer; **ocean pH decreases** / ocean acidification occurs
- Hard **corals cannot absorb calcium carbonate** they need to maintain their **skeletons**, stony skeletons that support corals will dissolve and corals destroyed / Coral polyp metabolism is affected and **corals expels the zooxanthellae**, (leaving the coral skeleton bleached), and eventual **death of corals due to lack of nutrients (provided by zooxanthellae)**

OR

- Photosynthesis in zooxanthellae is disrupted** at higher than usual temperatures, thus producing an excess of **products that are toxic**

2. Coral polyp metabolism is affected and **corals expels the zooxanthellae**, (leaving the coral skeleton bleached), and eventual **death of corals due to lack of nutrients (provided by zooxanthellae)**

[Total: 11]

Section B

Answer one question only.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts (a) and (b), as indicated in the question.

QUESTION 5

Cellulose is the most abundant biopolymer on earth. It forms a significant proportion of the dry mass of plants.

Outline the role of atmospheric CO₂ in contributing to the structure of cellulose in plants. [15]

Process of Calvin cycle

1. ref. **Calvin cycle** / light-independent reactions in **stroma** of **chloroplast**;
2. **CO₂** (1-carbon molecule) is **fixed** by combining with (a 5C compound called) **ribulose 1,5-bisphosphate (RuBP)**;
3. catalysed by the enzyme **rubisco (RuBP carboxylase)**,
4. to form an **unstable intermediate 6C compound**, which immediately splits into half to form **2 molecules** of (a 3C compound called) **glycerate-3-phosphate (PGA/GP)**;
5. **PGA/GP** is **reduced** to **triose phosphate (PGAL/TP)**;
6. The **electrons (hydrogen)** for this **reduction** come from **NADPH** produced from the **light-dependent reactions**
7. and the **energy** for this step comes from **ATP** produced from the **light-dependent reactions**;
8. For every **3** molecules of **CO₂**, there are **6** molecules of **triose phosphate / glyceraldehyde-3-phosphate / TP / PGAL / G3P**;
9. but only **1** molecule of **PGAL/TP (3C)** exits the cycle, and will be used by the plant cell to synthesise **carbohydrate** like glucose (sugar).
10. The other **5** molecules of **PGAL/TP** must be **recycled** to **regenerate** 3 molecules of **RuBP**;
11. RuBP is now prepared to receive **CO₂** again, and the cycle **continues**;

Max 8 pts

Structure of cellulose

12. ref. cellulose as a polymer of **β-glucose**;
13. Each β-glucose comes from **2 molecules of PGAL/TP** that exited Calvin cycle;
14. **6** carbon dioxide molecules required for synthesis of **1** molecule of β-glucose;

- 15. ref. **β (1→4) glycosidic bonds** between β -glucose molecules
- 16. catalyzed by the enzyme cellulose synthase.
- 17. adjacent units are oriented 180° to each other;
- 18. ref. cellulose exist as linear chains/form;
- 19. **Hydrogen bonds form between parallel chains** (as –OH points outwards) to establish cross-linkages between chains ;
- 20. ref. Microfibrils (and macrofibril) formation through association of many cellulose molecules (i.e. Many microfibrils combine to form macrofibrils)

Max 6 pts

QwC: [1mark] Clear, organised flow without ambiguity for Calvin cycle description and to include the link between cellulose (formed from β -glucose) and Calvin cycle.

QUESTION 6

Without genetic variation, some of the basic mechanisms bringing about evolutionary change cannot operate.

Explain how mutation events may contribute to variation, and how this variation can influence natural selection. [15]

1. ref. **germ-line mutations** / mutations that occur in the **gametes**, being **inherited** by the offspring;
2. ref. phenotype (characteristics) of the organism;

Gene mutation [max 4]

3. Spontaneous errors made during **DNA replication / DNA recombination / DNA repair**
4. ref. a change in the nucleotide / base sequence (may involve one or more bases) of the DNA molecule in a particular/single gene locus / region on a chromosome
5. ref. insertion/deletion/substitution/inversion of incorrect nucleotides
6. ref. an alteration in the DNA sequence may change the amino acid sequence
7. ref. introduction of **new alleles** into gene pool;

Chromosomal aberration [max 6]

8. Several gene loci are involved
9. Involves reshuffling of existing alleles in the gene pool (no creation of new alleles)

Change in the number of chromosomes

- 10.ref. non-disjunction in mitosis or meiosis
- 11.ref. failure of sister chromatids to separate during anaphase in mitosis / homologous chromosomes to separate during anaphase I in meiosis / chromatids to separate during anaphase II in meiosis
12. Can affect one, several or all the chromosomes within a nucleus
13. leads to a **change in the number of chromosomes**

Change in structure of chromosomes

- 14.ref. chromosomal breaks that occur during mitosis or meiosis
15. resulting in deletion, duplication, inversion or translocation of chromosomal segments
16. leads to a **change in the structure of chromosomes**
- 17.ref. reshuffling of alleles on the affected chromosomes

Natural Selection [max 3]

- 18.ref. different selection pressures which select for / favour individuals with selective advantage / beneficial phenotype/alleles;
19. More survive till reproductive age/maturity and reproduce to pass on advantageous alleles to their offspring;

20. After many generations / over time, natural selection will lead to change in frequency of alleles.

QwC: [1mark] Clear, organised flow without ambiguity for explanation of evolution with natural selection, and to include at least 1 description for gene mutation and chromosomal aberration;

- END OF PAPER -